

Benjamin Menetrier

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Positions

- **May 2023 - present:** Researcher at the Norwegian Meteorological Institute (Oslo, Norway).
- **October 2018 - April 2023:** Research scientist at IRIT (Toulouse), subcontracting for the JCSDA (Boulder, CO). Main developer of the SABER library in the JEDI project.
- **April 2016 - September 2018:** Research scientist at CNRM (Toulouse).
- **October 2015 - March 2016:** Research engineer at CERFACS (Toulouse).
- **Septembre 2014 - September 2015:** Project Scientist I at NCAR (Boulder, CO).

Education

- **2011-2014:** PhD at CNRM (Toulouse): "Using ensemble data assimilation for the modelling of flow-dependent background error covariances at convective scale", Léopold Escande award 2014.
- **2011:** ENS (Paris) graduation, geosciences specialty.
- **2010:** Master "Ocean, Atmosphere, Climate and Remote Sensing" at UPMC (Paris).

Skills

- Language: fluent English, French mother tongue.
- Good knowledge of Linux, shell scripting, word-processing with Latex.
- Extensive knowledge of Fortran, good working knowledge of C++ and Python.
- Long experience as a HPC user on many different platforms.
- Developer in large, complex and distributed-memory codes of data assimilation (IFS/ARPEGE-AROME, WRF-DA, GSI, NEMOVAR, OOPS, JEDI framework).

Teaching

- **2016 - 2025:** training about ensemble-variational methods for data assimilation at CERFACS (Toulouse).
- **2024:** lecture and practical training for developers on the OOPS system (ACCORD consortium).
- **2019 - 2023:** lectures and practical training for the "JEDI academies" organised by the JCSDA.

Bibliometry and conferences

- Author of 19 published papers in international peer-reviewed journals.
- Cited 424 times, H-index of 12 [Web of Science, 07/01/2026].
- ORCID account: 0000-0002-2593-3398 / Web of Science account: ABG-7190-2021

Selected recent papers

- **Ménétrier, B.** (2026): The Normalized Interpolated Convolution from an Adaptive Subgrid (NICAS) method, EGUsphere, doi: 10.5194/egusphere-2025-5780 (preprint).
- Jung *et al.* (2024): Three-dimensional variational assimilation with a multivariate background error covariance for the Model for Prediction Across Scales - Atmosphere with the Joint Effort for Data assimilation Integration (JEDI-MPAS 2.0.0-beta), Geosci. Model Dev., 17, 3879-3895.
- Liu *et al.* (2022): Data assimilation for the Model for Prediction Across Scales - Atmosphere with the Joint Effort for Data assimilation Integration (JEDI-MPAS 1.0.0): EnVar implementation and evaluation, Geosci. Model Dev., 15, 7859-7878.
- Laloyaux, P., Frolov, S., **Ménétrier, B.**, and Bonavita, M. (2018). Implicit and explicit cross-correlations in coupled data assimilation. *Q J R Meteorol Soc.*, 144, 1851-1863.
- Auligné, T., **Ménétrier, B.**, Lorenc, A. C., and Buehner, M. (2016). Ensemble-Variational Integrated Localized Data Assimilation. *Monthly Weather Review*, 144, 3677-3696.
- **Ménétrier, B.**, and Auligné, T. (2015). Objective optimization of localized-hybridized ensemble-based covariances. *Monthly Weather Review*, 143(10), 3931-3947.